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AI Applications in Medical Reporting and Diagnosis

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***Abstract*—The integration of artificial intelligence (AI) in healthcare is revolutionizing diagnosis and patient care by improving clinical documentation and the management of electronic health records that depend on medical image interpretation, increasing accuracy, and reducing time. Egypt ranks first in liver disease and second in liver cancer mortality worldwide in 2020. Large language models, a subset of AI techniques, can assist in disease diagnosis. LLM models with multimodal capabilities can classify and describe patient scan images and extract information from clinical notes. These models can extract vital diagnoses with the support of prompt engineering, as one of these models can answer questions, summarize information, and translate complex medical terminology into plain language, enabling patients to understand their medical reports and diagnoses. There are two primary approaches to achieving this. First, fine-tuning can adapt the model to medical data, which can be resource-intensive. The second approach, pre-trained LLM models can be utilized to leverage pre-trained models to perform the necessary tasks, focusing on effectively using prompts to guide the model for precise and relevant outputs. This study highlights the role of generative AI models by focusing on prompt engineering, and how carefully crafting prompts can enhance the effectiveness of LLM models in medical applications with high accuracy. It demonstrates this through experiments using pre-trained models based on semantic similarity with GPT-4o and BioGPT. Implementing a zero-shot model for liver tumor classification is one of the prompt engineering techniques. The performance metrics achieved were impressive, accuracy, precision, recall, and F1-scores are 88, 81, 88, and 83 percent, respectively.**

***Index Terms*—Generative AI, Prompt Engineering, Large Language Model (LLM), ChatGPT, Artificial Intelligence, Medical radiology reports**

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I. Introduction

Generative AI can be implemented in prompt engineering as a cutting-edge approach in natural language processing (NLP) tasks by improving LLM performance [1], [2]. The healthcare sector has recently received increasing attention due to its close relationship with each individual. In generative AI models, prompt engineering is user-provided textual input that directs the model's output, simple queries to a complex problem. Prompt engineering is a computer program that can simulate conversations with human users. The generative AI applications in healthcare and medicine will grow in significance as they develop and become more precisely adapted to the particular conditions and demands of the medical domain. This is especially helpful when training data is lacking or certain jobs are too specialized for broad models, such as [3]–[6]. Factors influencing prompt criteria are user intent, model understanding, domain specificity, clarity, specificity, and constraints [7]. In artificial intelligence (AI), natural language processing (NLP) is a subfield that can transform unstructured clinical notes into structured data. It can also be used to identify medical conditions, medications, and names of lab tests through tasks like named entity recognition (NER). GPT models are a notable innovation in NLP that was first introduced by OpenAI [3].

Large transformers, known as LLMs are composed of either an encoder, a decoder, or both and are trained in unsupervised learning on a large corpus of text. One of the earliest examples of this type of transformer is BERT, an encoder transformer trained in unsupervised learning to predict masked tokens. Next-token prediction was utilized by decoder-style transformers such as GPT. These models have significantly advanced various downstream tasks, in-

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cluding question answering, code completion, and summarizing. Several improvements in this model led to models like GPT-3, GPT-3.5, and most recently GPT-4 (OpenAI 2023) [8]. Prompt engineering has demonstrated significant superiority across multiple areas and has gained major importance in healthcare because of recent improvements in LLM models. LLMs can connect radiation oncology data to clinical applications. Currently, oncologists often have to spend considerable time reviewing extensive clinical notes, which include a variety of text documents such as demographics, operation notes, pathology and radiology reports, lab results, discharge, and consult notes stored in the Electronic Medical Record (EMR) systems [9]. LLM models like ChatGPT can continue evolving and improving medical applications, such as radiology reports, adding value and contributing to human health. Despite this, LLM models have many limitations, including transient state, probabilistic nature, outdated information, content fabrication, resource intensity, and domain specificity. So, prompt engineering techniques enhance LLM utility and mitigate inherent constraints. Prompt engineering has more powerful tools like a chain of thought prompting, encouraging the model to be factual through other means, explicitly ending the prompt instructions, using the AI to correct itself, generating different opinions, teaching an algorithm in the prompt, and affordances [10]. Prompt engineering optimizes the interaction with the chatbot for exact findings. Prompt engineering is divided into manual prompts (zero-shot, one-shot, few-shot, embedding-based retrieval mechanisms) and automated prompts (discrete, continuous). Manual prompts depend on human-crafted prompts. A prompt that combines instructions, questions, input data, and examples [1], [7], [10]. Automated prompts are generated using algorithms or automating methods that can save time and cost which collect task-specific information. Prompt engineering has many applications, including classification, generation, detection, augmentation, question answering, inference, Zero-shot learning, and embedding-based retrieval mechanisms. Prompt engineering can help simplify radiological reports, extract information from oncology notes, and classify fractures using free-text reports even with a limited dataset [5], [11]–[13]. Also, image segmentation forms the basis of various clinical tasks, including surgical planning, diagnostic assistance, and surgical navigation. Segmentation can be classified as Atlas-based, learning-based methods, and vision foundation models [14]. This paper delves into the development and methodology of innovative AI tools, aiming to address the challenges posed by explicit content in radiology reports. The integration of GPT-4, a state-of-the-art generative language model, plays a pivotal role in categorizing liver tumors. Section 2 demonstrates related work highlighting the effectiveness of prompt engineering across various tasks, showcasing its potential to enhance large language model (LLM) performance in medical contexts by building on AI techniques. Next, section 3 shows the methodology of prompt engineering for this work. After that, section 4 shows the experimental results and evaluation of the proposed prompt engineering. Finally, section 5 provides a conclusion and future work of the paper.

II. Related Work

Deep learning have been applied in varieties of application domains [15]–[18]. Among those domain, Medical doamin [19]–[22]. Particularly, several research efforts used LLM for medical domains. For example, the authors in [23] discussed Llama 3. The model is a dense transformer with 405B parameters and a context window of up to 128K tokens. Llama 3 integrates image, video, and speech capabilities with a compositional approach. Another research study [12] used clinical radiology cases to concatenate prompting strategies with LLM ChatGPT and GPT4. Also, this paper [24] used GPT-4, GPT-3.5-turbo, Gemma-7B and Llama3-8B. The authors employed prompt engineering to boost accuracy. GPT-4 achieved the highest accuracy in detecting disease status. The work in [11] explains that the study introduces a prompt construction module for medical language templates that enables extract of contextual information from electronic health patient records and generates complete cell embedding. The authors in [25] demonstrated that AI models can accurately segment liver tumors without manual annotation by using synthetic tumors in CT scans. The work in [14] concluded with the following contributions. Firstly, the introduction of the SAM prompt engineering framework. Secondly, the development of two efficient image registration strategies that align images or prompts with the referenced dataset from the Royal National Orthopaedic Hospital. Third, the demonstration of improved performance compared to alternative approaches which require full segmentation of the reference set and using SAM without the proposed registration strategy, for clinical image segmentation. So, the prompt engineering approach for segmentation foundation models depends on matching the input image or prompt to a small dataset of weakly labeled samples. This alignment allows the construction of effective prompts for segmentation foundation models to obtain accurate segmentation, even for out-of-distribution samples, using as few

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as five-point prompt-labeled reference images. The experiments for the BERT and BART-based models in work in [13] were conducted by comparing the performance of BERT-based models and prompt engineering on LLM. Every model was tuned for seven epochs, starting at a learning rate of 2e-05, 128 was the batch size, and 0.01 was the weight decay parameter. The Command-R and LLAMA3 models showed significantly better performance in their results. Authors in [26], hypothesize that applying in-context learning from text-based models to medical images could be immensely beneficial. This approach is particularly relevant for underrepresented medical cases, such as rare tumor types, which often receive insufficient representation in traditional deep-learning training pipelines. To explore this, they conducted experiments using the GPT-4V model through the official OpenAI Python API's chat completions endpoint, testing the model across various datasets. The reported performance metrics include an accuracy score of 82.7% for GPT-4V compared to 83.3% on the MHIST dataset and 92.7% compared to 88.3% on the PatchCamelyon dataset. These experiments used a zero-shot baseline as a comparison to evaluate the benefits of in-context learning. The authors have laid the groundwork for a general-purpose framework that advances state-of-the-art prompting techniques for image analysis.

III. Methodology

This paper employs prompt engineering architectures for dealing with medical reports. The proposed method utilized the https://www.eurorad.org dataset for experiments and evaluation. The dataset comprises a sample of multi-modality radiology reports that contain text and CT or MRI scans. The dataset was designed to represent a sample of medical reports containing clinical findings, associated clinical notes, and diagnostic details. Prompt engineering can solve The complex and diverse professional terminology in the medical field, the wide range of professional knowledge, and ethical issues related to patient privacy in the healthcare field [11]. In this paper, we fine-tuned BioBERT for medical reports using the GPT-4 model. Cosine similarities are calculated with encoded dialogues from the training data to summarize each dialogue, where k=3 with GPT-4 along with section headers to get a summary for the given part. The value of k fits well with the context length of the prompting-based pipeline [27], [28]. Accuracy was calculated as the proportion of correct predictions to the total predicted numbers in Eq.1. The accuracy steadily improves over the training epochs, reflecting the models' increasing ability to correct interpret, and classify medical texts. Precision and recall metrics revealed the balance between false positives and false negatives, crucial for applications in clinical decision-making where the cost of errors varies significantly Eq.2, 3. The F1-score [29] highlighted the overall effectiveness of the models, considering both precision and recall and showed consistent improvement, suggesting that the models were well-tuned to handle the complexities of the medical datasets Eq.4.

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{3}$$

$$F1 - Score = \frac{2(Precision)(Recall)}{(Precision) + (Recall)} \tag{4}$$

A. Problem Identification and Motivation :

Recognizing the need for radiology-friendly versions of medical radiology reports, our research addresses the challenge of categorizing liver tumors in patient reports accurately.

B. Objectives of the Design :

Develop an innovative tool that utilizes advanced AI techniques to detect whether the patient is normal or abnormal systematically and the type of tumor if present.

C. Design and Development

We use the pre-trained model to perform the desired task. Prompt engineering can be broken down into five key steps. First, we use an appropriate pre-trained model such as GPT-4o and BioGPT, suitable for generating radiology reports. Next, design the prompts for downstream tasks, which can be tailored to generate radiology reports; this step is called the prompt engineering process. The third step involves designing responses based on the task at hand, allowing the model to produce the desired output. In the fourth step, we expand the paradigm to improve results or adaptability methods. Finally, researchers must design training strategies that enable the model to learn efficiently and effectively, save time, and improve accuracy in classifying medical texts or answering questions related to specific cases. Fig. 1 explains the following steps:

1) Medical Reports Loading: load the radiology reports into memory using a Python library such as Pandas or Pydicom, depending on the format of the reports. That establishes a starting point for the analysis process by importing the raw report data.

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2) Text preprocessing: utilize natural language processing (NLP) techniques to clean and standardize the text data from the reports that define keywords related to liver conditions such as liver, hepatic, tumor, lesion, abscess, carcinoma, metastasis, cirrhosis, and foreign object. Then prepare the text for further analysis by removing noise, such as specific unwanted patterns (like dates, DOIs, ISSNs, author names, etc.). Standardize formatting, and handling common abbreviations, is done by splitting the cleaned text into smaller, relevant sentences.

3) Named Entity Recognition (NER): apply a medical (NER) model, such as one based on BERT or BioBERT to identify and extract key medical entities from the report text to isolate important medical terms, anatomical structures, and findings for focused analysis.

4) Information Extraction: that develop rule-based or machine-learning algorithms to extract specific information such as diagnoses, measurements, and recommendations from the identified entities and surrounding context. The result is to structure the key information from the reports for easier analysis and comparison.

5) Report Categorization: implement a classification model, such as a fine-tuned BERT or a traditional machine learning classifier, to categorize reports based on their content (normal condition, haemangioma, benign, focal fatty sparing, cirrhosis, hepatocellular carcinoma (HCC), liver tissue appears ablated, presence of liver metastasis, liver abscess identified, foreign body). As a result, reports are organized into meaningful categories to facilitate prioritization and further analysis.

6) Data Visualization: generate visual representations of the extracted data using libraries like Matplotlib or Plotly in Python. This creates easily interpretable summaries of the report findings for quick review by radiologists or referring physicians.

7) Integration with PACS/RIS: develop interfaces to integrate the processed report data with existing Picture Archiving and Communication Systems (PACS) or Radiology Information Systems (RIS). To ensure seamless incorporation of the analyzed data into existing clinical workflows.

D. Evaluation

A comprehensive evaluation will be conducted with a panel of experts in medical informatics and AI technology to optimize the benefits of prompt engineering in processing medical reports. The evaluation process focuses on the accuracy of information extraction that can determine and evaluate whether the extracted and categorized information retains

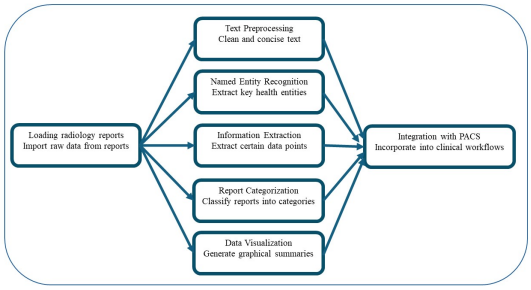

Fig. 1: Radiology report processing workflow

the clinical significance of the original reports. Measure the accuracy of automated report categorization compared to expert manual classification. Evaluate the ease of integrating existing electronic health record (EHR) systems. Assess the accessibility and usability of the processed information within current clinical workflows. The results of this comprehensive evaluation will guide the refinement and enhancement of the medical report processing system, ensuring it meets the high standards required for clinical use. By providing an efficient, accurate, and reliable tool for managing medical reports, the goal is to improve patient care and positively contribute to medical research. This system can work on diagnosing other diseases.

1) Model Initialization: The script begins by initializing a pre-trained model (BioGPT) and its tokenizer using Hugging Face's transformers library, the OpenAI API with the ChatGPT model. BioGPT is a model specifically designed for biomedical text generation tasks.

2) Data preparation: The training and evaluation datasets are loaded from JSON files in Google Drive. The datasets are tokenized and prepared using a custom class (Radiology Dataset) to ensure compatibility with the model's input requirements.

3) Training Configuration: The training arguments object defines the parameters for training, such as output directory, number of epochs, and batch size.

4) Model Training and Evaluation: A trainer object is created to handle the model training process, incorporating the defined arguments, datasets, and tokenizers.

5) Saving the Trained Model: The trained model is saved to the specified output directory for future use. This methodology section integrates various prompt engineering techniques to handle complex medical reports. The following section presents the experimental results, where we evaluate the effectiveness of our proposed methodology using a comprehensive set of experiments and performance metrics

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IV. Results

In our preliminary evaluation of the AI tools, we conducted a self-evaluation using a sample of medical reports. The study focused on the contributions of prompt engineering to healthcare natural language processing applications, including question-answering systems, text summarization, machine translation, and tumor classification. AI techniques demonstrated the capability to adapt to new scenarios with minimal labeled data [11]. Our experiments involving prompt selection via semantic similarity with GPT4 yielded the most favorable outcomes on the validation split. Prompt selection proved the most effective approach for medical reports and GPT4. The performance of prompt selection is attributed to the in-context medical reports selected for semantic similarity with the input dialogue. This approach generated coherent and accurate summaries.

A. Experiment 1 (Zero-Shot classification)

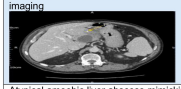
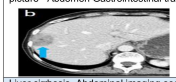
cases	Diagnosis from radiology report	Prompt Engineering zero-shot-classification results	
		Classification	Score
Adrenal infarction and hemorrhage: Diagnosis with CT scan - Bilateral adrenal infarction and unilateral adrenal hemorrhage	Bilateral adrenal infarction and unilateral adrenal hemorrhage	Bilateral adrenal infarction and unilateral adrenal hemorrhage	0.7238
		Focal fatty sparing in the liver	0.4871
		Liver abscess	0.3364
		Cirrhosis of the liver	0.3227
		Liver metastasis	0.2789
		Ablated liver tissue	0.2525
		Hepatocellular carcinoma (HCC)	0.2334
		Liver Hemangioma	0.2038
		Benign liver lesion	0.1576
		Normal liver condition	0.0606
Dropped Gallstone as an Uncommon Cause of Pleural Empyema and Perihepatic Abscess	Perihepatic abscess caused by a dropped gallstone associated	Liver metastasis	0.5133
		Liver abscess	0.5034
		Focal fatty sparing in the liver	0.4426
		Bilateral adrenal infarction and unilateral adrenal hemorrhage	0.4085
		Ablated liver tissue	0.3852
		Cirrhosis of the liver	0.3763
		Hepatocellular carcinoma (HCC)	0.3675
		Liver hemangioma	0.3578
		Normal liver condition	0.0487
		Benign liver lesion	0.0378
A fishy tale: Foreign body ingestion resulting in hepatic abscess formation- Abdominal imaging 	Hepatic Abscess Secondary to Gastric Perforation by Foreign Body	Liver tissue appears ablated	0.9644
		Liver abscess due to foreign body	0.9534
		Liver metastasis detected imaging	0.8915
		Cirrhosis of the liver detected	0.8794
		Focal fatty sparing in the liver	0.8620
		Liver shows signs of hemangioma	0.8155
		Hepatocellular carcinoma (HCC)	0.5490
		Benign liver lesion identified	0.2939
		Normal liver condition	0.0959
Atypical amoebic liver abscess mimicking liver malignancy – the importance of seeing the full picture - Abdomen Gastrointestinal tract Liver 	Amoebic liver abscess and amoebic colitis	Liver abscess	0.9374
		Focal fatty sparing in the liver	0.8620
		Liver metastasis	0.8618
		Ablated liver tissue	0.8531
		Liver hemangioma	0.7025
		Cirrhosis of the liver	0.6650
		Hepatocellular carcinoma (HCC)	0.5681
		Benign liver lesion	0.3244
		Normal liver condition	0.0959
Liver cirrhosis- Abdominal imaging section	Macronodular liver cirrhosis	Cirrhosis of the liver	0.9673
		Liver hemangioma	0.8689
		Ablated liver tissue	0.8493
		Hepatocellular carcinoma	0.8279
		Liver abscess, Score	0.7461
		Liver metastasis	0.4939
		Focal fatty sparing in the liver	0.3025
		Normal liver condition	0.0758
		Benign liver lesion	0.0291
A bridge between the liver and the Heart - Cirrhosis Neoplasia Case Type	Hepatocellular carcinoma (HCC) with inferior vena cava involvement	Liver metastasis	0.9496
		Hepatocellular carcinoma (HCC)	0.9480
		Cirrhosis of the liver	0.9322
		Liver hemangioma	0.9250
		Liver abscess	0.9235
		Ablated liver tissue	0.9150
		Focal fatty sparing in the liver	0.8112
		Bilateral adrenal infarction and unilateral adrenal hemorrhage	0.7748
		Normal liver condition	0.2137
		Benign liver lesion	0.1056

Fig. 2: The integration of AI technology into traditional radiology reports

To classify liver conditions from radiology reports containing CT or MRI scans and unstructured doctor notes, we utilized a zero-shot classification model using LLM specifically the 'Facebook/bart-large-mnli' model from the Hugging Face transformers library. We crafted specific prompts based on extracting the medical report content, focusing on key clinical indicators and findings. Fig. 2 present a comparative analysis between radiological diagnoses and AI/ML-based classification outcomes with a high accuracy score in diagnosing liver tumors. The left column shows different medical cases, each accompanied by an image from the diagnostic imaging in CT scans or MRIs with a brief description of the results from the radiology report. These results demonstrate how closely the AI model's predictions align with the expert radiologist's diagnosis, providing a quantitative measure of the model's accuracy and reliability in interpreting radiological scans. The figure effectively highlights both the potential and limitations of AI in detecting diagnostic processes, providing insights into areas where prompt engineering models perform well and where further refinement might be needed to match human expert accuracy. The multi-label=True option allows the model to classify the text under multiple categories simultaneously, essential in medical contexts where a patient may have more than one condition Table 1.

TABLE 1: Prompt Engineering for Liver Tumor Classification

Aspect	Description
Objective	Classify liver conditions based on extracted text using a zero-shot classification model.
Process	1. Extract text from a PDF document using PyPDF2. 2. Use a zero-shot classification model with the "facebook/bart-large-mnli" model to analyze the text. 3. Classify the text into potential liver conditions based on pre-defined labels.
Labels	Normal liver condition • Liver haemangioma • Benign liver lesion • Focal fatty sparing in the liver • Cirrhosis of the liver • Hepatocellular carcinoma (HCC) • Ablated liver tissue • Liver metastasis • Bilateral adrenal infarction and unilateral adrenal hemorrhage • Liver abscess
Output	The model provides classifications with corresponding scores, indicating the Possibility of each condition based on the extracted text.

B. Experiment 2 (Rule-Based Prompt Engineering)

Analyze the medical report to determine evidence of a hyperdense subcapsular rounded lesion in the input medical report. If this keyword is found, it triggers a specific interpretation in the output, suggesting the possibility of a tumor, with additional context about post-TACE changes as in Table 2.

C. Experiment 3 (Automatic Prompt Engineering)

Radiology reports contain critical information necessary for diagnosing and managing various medical conditions. The interpretation of these PDF reports often involves identifying specific findings,

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TABLE 2: Analysis of Medical Report

Aspect	Content
Prompt Engineering Code	Analyze the medical report to determine if there is evidence of a tumor or other significant findings. If "hyperdense subcapsular rounded lesion" is present in the report, it may indicate a tumor but corresponds with post-TACE changes. Additional findings include: • Shrunken cirrhotic liver with nodular surface. • No evidence of residual recurrent lesions of abnormal contrast enhancement. • Chronic partially recanalized portal vein thrombosis. • Splenic artery aneurysm noted.
Tumor Analysis Result	There is a hyperdense subcapsular rounded lesion, which could be indicative of a tumor, but it corresponds with post-TACE changes, suggesting it has been adequately treated.

measurements, and disease states. This is done by extracting key clinical findings using regular expressions (regex) and generating structured prompts for clinicians to review. The methodology is done by reading PDF reports and converting the unstructured text into a single string for programmatic processing. Use regex to identify and extract key terms related to liver conditions such as lesion location, steatosis status, cirrhosis status, lesion size, and liver tumor status. Based on the extracted findings, the script generates two types of prompts: Discrete Prompts: Request the clinician to verify specific findings that can address categorical information such as the presence or absence of a tumor as in Table 3. Continuous Prompts: Focus on measurements, such as confirming lesion size and tracking changes over time. They facilitate consistent documentation of quantitative data.

TABLE 3: Automatic Prompt Engineering

Discrete Radiology Prompts	Lesion Location	Segmented. Verify location and check for any additional lesions.
	Check hepatic steatosis status	Unknown
	Cirrhosis findings	None. Please confirm or provide additional details. Liver tumor and cell conditions: Hepatocellular carcinoma. Please verify the details.
	Liver tumor and cell conditions.	Hepatocellular carcinoma. Please verify the details.
Continuous Radiology Prompts	Confirm the size of the lesion is segmented. The current measurement is approximately 4.1 cm. Measure any changes in lesion size from the previous scan. The current size is 4.1 cm.	

D. Experiment 4 (Fun-tuning prompt engineering)

In this experiment, we evaluate the performance using the two advanced medical language models (BioGPT and BioBERT v1.1 (NER)) on the sample of the medical radiology reports dataset as in Table 7. The evaluation was conducted through many steps, including data preparation, model fine-tuning, and performance measurement across multiple metrics.

1) Installation of Libraries and Preparation of Data: We began by installing the necessary libraries, transformers, and sacremoses, to facilitate the use of pre-trained language models and tokenization processes. This dataset was structured and saved in JSON format to ensure compatibility with the models and ease of use in subsequent training phases.

2) Model and tokenizer upload: Tokenizers specific to the medical domain were uploaded to preprocess the text data effectively. We utilized the tokenizers corresponding to BioGPT, BioBERT v1.1 (NER). These models were chosen due to their fine-tuning of extensive biomedical literature, enabling them to handle complex medical language effectively.

3) Dataset preparation: The radiology datasets were loaded and divided into training and evaluation subsets.

4) Performance Evaluation Metrics: The models were fine-tuned using the prepared training dataset, and their performance was evaluated on the validation set using standard metrics.

5) Results Storage: The computed metrics were systematically saved in the 'Radiology Model Results' directory as shown in Table 4.

6) Visualization of Results: Facilitating the interpretation of the model's performance, we plotted the evaluation metrics across epochs. These visualizations provided insights into the learning dynamics and stability of the models over time, as in Fig. 3 and Table 5.

TABLE 4: Model Evaluation Summary

Metric	Value
Evaluation Loss	1.4798
Evaluation Accuracy	0.7205
Evaluation Precision	0.6514
Evaluation Recall	0.7205
Evaluation F1-Score	0.6701
Evaluation Runtime (s)	3.209
Evaluation Samples per Second	3.428
Evaluation Steps per Second	3.428
Epoch	50

TABLE 5: Performance Metrics for Medical Diagnosis Classes

Class	Precision	Recall	F1-Score	Support
Amoebic liver abscess and amoebic colitis	1.00	1.00	1.00	1
Bilateral adrenal infarction and unilateral adrenal hemorrhage	0.00	0.00	0.00	1
Ciliated hepatic foregut cyst	1.00	1.00	1.00	1
Common hepatic duct metastasis mimicking hilar cholangiocarcinoma	1.00	1.00	1.00	1
Hepatic abscess secondary to gastric perforation by foreign body	1.00	1.00	1.00	1
Liver abscess as a secondary complication of duodenal perforation	1.00	1.00	1.00	1
Metastatic hepatocellular carcinoma	0.50	1.00	0.67	1
Perihepatic abscess caused by a dropped gallstone associated with empyema	1.00	1.00	1.00	1
Accuracy	0.88 (8 samples)			
Macro Avg	0.81	0.88	0.83	8
Weighted Avg	0.81	0.88	0.83	8

Prompt engineering helps doctors and healthcare professionals quickly grasp the key information about a patient's condition without reading lengthy reports, which is done by medical Text Summarization. Also, we employed prompt engineering techniques using a pre-trained language model to enhance its performance in a question-answering task related to medical radiology as in Table 6.

TABLE 6: Results of Question Answering Model

Question	Context	Generated Answer
What is the patient's age?	The patient is a 71-year-old male with a history of liver disease.	The patient is a 71-year-old male with a history of liver disease and a patient with a history of liver disease but

Another tool is medical classification; we classify medical radiology reports into specific categories. After model selection and configuration, we utilized a custom-trained causal language model. This is done

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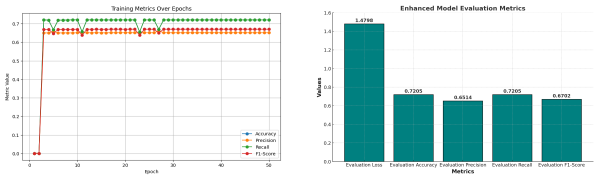
TABLE 7: Sample of https://www.eurorad.org for abdominal CT reports

Title	Area of Interest	Imaging Technique	Special Focus
Common hepatic duct metastasis mimicking hilar cholangiocarcinoma	Abdomen Biliary Tract / Gallbladder Liver	CT, MR, MR-Cholangiography	Metastases
A fishy tale: Foreign body ingestion resulting in hepatic abscess formation	Abdomen	CT, MR	Abscess Foreign bodies
Atypical amoebic liver abscess mimicking liver malignancy	Abdomen Gastrointestinal tract Liver	CT, MR	Abscess Infection Tropical diseases
Dropped Gallstone as an Uncommon Cause of Pleural Empyema and Perihepatic Abscess	Biliary Tract/Gallbladder Thorax	CT	Abscess Acute
Ciliated hepatic foregut cys	Abdomen Gastrointestinal tract Liver	Ultrasound,Ultrasound -Colour Doppler, MR	Congenital Cysts
Adrenal infarction and hemorrhage: Diagnosis with CT scan	Abdomen Anatomy	CT, CT-Angiography	Haemorrhage Ischaemia/Infarction
Liver abscess as a consequence of duodenal perforation by fishbone	Abdomen Emergency Gastrointestinal tract Liver Small bowel	CT, Ultrasound	Abscess Foreign bodies Infection
Hepatocellular carcinoma presents as a pancreatic metastasis with no primary liver tumour	Abdomen Pancreas	CT, MR, PET-MR	Cancer

by fine-tuning radiology-related data to ensure the model's relevance to the domain-specific context. In this step the model and tokenizer were loaded from a local path to optimize computational efficiency, the model was configured to run on a GPU, if available; otherwise, it defaulted to using the CPU.



(a) Training matrices over epochs (b) Enhanced Model Evaluation Metrics

Fig. 3: visualize results

TABLE 8: Results of generated answer

Question	Context	Generated Answer
Medical record classification	Model: generate, max_new_tokens=1	MRI shows multiple lesions in the liver with irregular borders

In table 8 the classification model was fine-tuned using the pre-trained model. Input preparation value is AutoTokenizer. The inference process value is MutoModelForCausalLM. The generated output was decoded to produce a human-readable classification result. There are many benefits, such as analyzing sentiment in medical reports or patient feedback to understand their responses to treatments or medications. Predictive text generation for medical reports using the model to generate possible medical texts based on initial findings as in Table 9.

TABLE 9: Sentiment Analysis

Question	Context	Sentiment Answer
Analyzing sentiment	Creating sentiment	The patient feels much better after new treatment

Named Entity Recognition (NER) for Medical Entities identifying medical entities like drug names, diseases, symptoms, and body parts in the medical texts as in Table 10.

TABLE 10: Named Entity Recognition

Variable	Code Snippet
Medical_text	Patient was given aspirin and diagnosed with pneumonia.
Inputs	tokenizer(medical_text, return_tensors="pt")
NER_outputs	model.generate(inputs, max_new_tokens=50)
NER_result	tokenizer.decode(ner_outputs[0], skip_special_tokens=True)

We utilized the BioGPT model, a pre-trained language model specifically designed for biomedical text generation, to produce clinically coherent narratives based on patient-doctor notes. The model effectively synthesizes detailed medical information, such as diagnoses and treatment plans. Parameters were optimized to maintain text quality and relevance, using temperature, top-k, and top-p sampling methods to balance accuracy and creativity in the generated content.

The models demonstrated consistent improvement in accuracy and precision across multiple medical reports. These results highlight the potential of integrating prompt engineering with LLMs for medical applications. In the concluding section, we discuss the implications of our findings and potential directions for future research.

V. Conclusion and future work

The evaluation of the AI tool for processing medical reports has demonstrated its potential to enhance clinical documentation through accurate information extraction, meaningful categorization, and effective visualization. While the tool shows promise in improving efficiency and supporting clinical decision-making, further refinement is necessary to address challenges related to consistency, handling complex cases, and seamless integration with existing healthcare systems. The use of AI, particularly through prompt engineering, could potentially reduce the workload of clinicians by saving time in reviewing and interpreting reports. However, these savings may vary depending on the complexity of cases and specific workflows in different healthcare settings. Continued development and iterative testing will be essential to fully realizing its benefits in enhancing patient care and advancing medical research. In the future, we can evaluate, debug, and monitor chains built by the LLM framework using the Lang Smith application before production [30]. Also, we anticipate significant advancements in Retrieval-Augmented Generation (RAG), AI agents, and Collaborative Robot Workstations (CREW). These developments are expected to enhance the capabilities

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of AI systems and their integration into medical applications. Additionally, increasing the dataset size by adding more medical reports will be essential. During the deployment stage, we can collaborate with expert radiologists to train them and enhance the system's performance.

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